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MILITARY SPECIFICATION

POWER SUPPLY SUBASSEMBLY: 10525745

1. SCOPE

1.1 This specification covers one type of power supply subassembly, a part of Gun Direction Computer: M18. The power supply is a circuit board consisting of various capacitors, diodes and a transistor used in the power supply circuits of the computer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Components Parts

FSC 1220

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DRAWINGS

U.S. Army Munitions Command

F10525745	Power Supply Subassembly

PACKAGING DATA SHEET

M14002	Data Sheet, Master Packaging

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The power supply subassembly shall be manufactured in accordance with Drawing F10525745 and drawings pertaining thereto.

3.2 General specifications.- The power supply subassembly shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The power supply subassembly shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the power supply subassembly for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the power supply subassembly shall not be damaged or adversely affected and shall meet the requirements of 3.4.

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3.3.2 Shock. - The power supply subassembly shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating. - The power supply subassembly shall meet the requirements of 3.4.5 and 3.4.6 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage. - The power supply subassembly shall meet the requirements of 3.4.1 through 3.4.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating. - The power supply subassembly shall meet the requirements of 3.4.6 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of +135°F.

3.3.4.2 Non-operating. - The power supply subassembly shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance. - The power supply subassembly shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97% ± 2%.

3.3.6 Salt spray. - The power supply subassembly shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 0.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance. - Unless otherwise specified the power supply subassembly shall meet the performance requirements of this specification at standard ambient temperature of between 60° and 90°F and at a barometric pressure of 28 to 32 inches of mercury with a RH up to 90%.

3.4.1 Transistor.

3.4.1.1 Direct-current gain. - The direct-current gain shall be 155 maximum and 63.3 minimum, when the collector current (I_C) is 4.0 ± 0.2 amperes (amps) and the collector-to-emitter voltage (V_{CE}) is between 0 and -1.75 volts, direct-current (Vdc).

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3.4.1.2 DC leakage current.

3.4.1.2.1 Collector-to-base voltage (V_{CB}) of -2.0 Vdc.- The leakage current (I_{CO}) shall be 0.3 milliamperes (ma) maximum when V_{CB} is -2.0 ± 0.2 Vdc.

3.4.1.2.2 Collector-to-base voltage (V_{CB}) of -55.0 Vdc.- The leakage current (I_{CO}) shall be 2.6 ma maximum, when V_{CB} is -55.0 ± 3.0 Vdc.

3.4.1.2.3 Emitter-to-base voltage (V_{EB}) of -10 Vdc.- The leakage current (I_{EO}) shall be 1.0 ma maximum, when V_{EB} is -10 ± 1.0 Vdc.

3.4.1.3 DC saturation voltage (V_{CE} Sat).- The DC saturation voltage (V_{CE} sat) shall be 0.75 Vdc maximum, when I_C is 10.0 ± 0.5 amps and I_B is 1.0 ± 0.2 amps.

3.4.2 Diodes.

3.4.2.1 DC forward voltage.- The forward voltage drop (V_F) across each diode shall be 1.2 Vdc maximum, when the forward current (I_F) is 100 ± 10 ma dc.

3.4.2.2 DC leakage current.- The leakage current (I_R) through each diode shall be less than 0.4 microamperes (ua) dc, when each diode is back-biased with 350 ± 20 Vdc.

3.4.3 Capacitors.

3.4.3.1 DC leakage current.- The maximum leakage current for the capacitors shall be as specified in Table I, when the capacitance voltages (V_C) indicated are applied across the appropriate capacitors.

TABLE - I

DC LEAKAGE CURRENT

VOLTAGE (V_C) *

Capacitor	Voltage	rms Voltage (ac)	Maximum DC Leakage current (ua)
C1	20.0 ± 1.0	0	150
C2	20.0 ± 1.0	0	150
C3	60.0 ± 3.0	0	150
C4	60.0 ± 3.0	0	150
C5	60.0 ± 3.0	0	150
C6	60.0 ± 3.0	0	150
C15	60.0 ± 3.0	0	150
C7	60.0 ± 3.0	0	100

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Capacitor	Voltage	rms Voltage (ac)	Maximum DC Leakage current (ua)
C8	60.0 ± 3.0	0	100
C9	60.0 ± 3.0	0	100
C14	10.0 ± 0.5	0	120
C13	10.0 ± 0.5	0	120

* V_C = Vac + Vdc

3.4.3.2 AC capacitance.- The capacitors shall exhibit the ac capacitances specified in Table II, for the nominal capacitance value when V_C is equal to 10.0 ± 0.5 Vdc and 1.02 Vac root mean square (rms) at a frequency of 120 ± 5 cycles per second (cps).

TABLE - II
AC CAPACITANCE

	OPERATING TEMPERATURES					
	Low Capacitance		Nominal Capacitance		High Capacitance	
CAPACITOR	MICROFARADS (MF)					
	MIN	MAX	MIN	MAX	MIN	MAX
C1	150	720	215	720	215	860
C2	150	720	215	720	215	860
C3	45	180	65	180	65	217
C4	45	180	65	180	65	217
C5	45	180	65	180	65	217
C6	45	180	65	180	65	217
C7	5.7	20.5	8.1	20.5	8.1	24.5
C8	5.7	20.5	8.1	20.5	8.1	24.5
C9	5.7	20.5	8.1	20.5	8.1	24.5
C10	0.054	0.154	0.064	0.154	0.064	0.17
C11	0.054	0.154	0.064	0.154	0.064	0.17
C12	0.018	0.052	0.021	0.152	0.021	0.059
C13	190	845	272	845	272	1015
C14	190	845	272	845	272	1015
C15	45	180	65	180	65	217

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3.4.4 Resistor.- The dc resistance across resistor, R1, shall be 0.1 ohm ± 15% with a current of no greater than 1.1 Adc (amperes, direct-current) flowing.

3.4.5 Low operating temperature (-40°F).

3.4.5.1 Transistor.

3.4.5.1.1 Direct-current gain.- The transistor shall meet the requirement of 3.4.1.1 except that the minimum gain shall be 53.9.

3.4.5.2 Diodes.

3.4.5.2.1 DC leakage current.- The diodes shall meet the requirements of 3.4.2.2.

3.4.5.3 Capacitors.

3.4.5.3.1 AC capacitance.- The capacitors shall meet the requirement of 3.4.3.2, excepting the AC capacitances indicated under the low capacitance column of Table II shall apply.

3.4.6 High operating temperature (+135°F).

3.4.6.1 Transistor.

3.4.6.1.1 DC leakage current.- The transistor shall meet the requirements of 3.4.1.2.2, except I_{CO} shall be 6.0 ma maximum.

3.4.6.2 Diodes.

3.4.6.2.1 DC leakage current.- The leakage current (I_R) of each diode shall be 4.0 ua maximum when each diode is back-biased with 350 ± 20 Vdc.

3.4.6.3 Capacitors.

3.4.6.3.1 AC capacitance.- The capacitors shall meet the requirement of 3.4.3.2, excepting the AC capacitances indicated under high capacitance column of Table II shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced in this specification and in the contract, shall constitute the minimum amount of inspection performed by the supplier prior to submission of the product for Government acceptance, and will constitute the maximum severity to be used by the Government for acceptance or rejection of any submitted lot. The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables III and IV and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table III shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE III

(100 percent inspection)

Characteristic	Requirement	Test Procedures
101. Transistor Direct Current Gain	3.4.1.1	4.5.2.1.1
102. DC leakage current collector to base (-2.0 Vdc)	3.4.1.2.1	4.5.2.1.2.1
103. DC leakage current collector to base (-55.0 Vdc)	3.4.1.2.2	4.5.2.1.2.2
104. DC leakage current Emitter to base (-10 Vdc)	3.4.1.2.3	4.5.2.1.2.3
105. Transistor DC Saturation Voltage	3.4.1.3	4.5.2.1.3

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Characteristic		Requirement	Test Procedures
106.	Dc forward voltage (diodes)	3.4.2.1	4.5.2.2.1
107.	Dc leakage current (diodes)	3.4.2.2	4.5.2.2.2
108.	Dc leakage current (capacitor)	3.4.3.1	4.5.2.3.1
109.	Ac capacitance	3.4.3.2	4.5.2.3.2
110.	Resistor (resistance)	3.4.4	4.5.2.4

4.3.1 General.- One Power Supply shall be selected at random as a control sample from each 50 produced. The control sample shall meet the requirements and tests in Table III.

CONTROL TEST		REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

4.3.2 Environmental.- Three units shall be selected at random as a control sample from each 100 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table III and shall then meet the requirements and tests in Table IV.

TABLE IV

CONTROL TEST		REQUIREMENT	TEST PROCEDURES
303.	Vibration	3.3.1	4.5.1.1
304.	Shock	3.3.2	4.5.1.2
305.	Humidity-Operating	3.3.4.1, 3.4.6	4.5.1.3.2
306.	Humidity-Non-operating	3.3.4.2	4.5.1.3.1
307.	Salt spray	3.3.6	4.5.1.4
308.	Transistor dc gain (-40°F)	3.4.5.1.1	4.5.2.5.1.1
309.	Diode dc leakage current (-40°F)	3.4.5.2.1, 3.4.2.2	4.5.2.5.2.1
310.	Capacitor-ac capacitance (-40°F)	3.4.5.3.1, 3.4.3.2	4.5.2.5.3.1
311.	Transistor dc leakage current (+135°F)	3.4.6.1.1, 3.4.1.2.2	4.5.2.6.1
312.	Diode dc leakage current (+135°F)	3.4.6.2.1	4.5.2.6.2
313.	Capacitor-ac capacitance (+135°F)	3.4.6.3.1	4.5.2.6.3
314.	Storage temperature	3.3.3.2	4.5.1.5

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4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- The test circuits shown in Figures I to 4 inclusive are given only as a guide. Any test circuit configuration that meets the requirements of 3.4 may be used. In conjunction with 4.4.1, the requirements of Tables IX to XI inclusive and the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5.

Item Number	Description
1. Voltmeter	D.C, 0-400 volt range; input impedance greater than 10 megohms
2. Dc Ammeter	Accuracy of not less than 2 percent of full range.
3. Signal generator	Frequency 0 to 1 KC. Distortion less than 3%.
4. Capacitance Bridge	Measurement accuracy ± 2%. Bridge sensitivity shall be such that capacitance measurements can be balanced to an accuracy of ± 0.2 percent.
5. Dc Ohmeter	Accuracy of not less than 1 percent.
6. Ac Ammeter	Accuracy of not less than 2 percent.

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4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration. - The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the power supply rigidly mounted in its normal operating position. Subsequent to the vibration test, the power supply shall show no damage or defect resulting from the test when examined by visual and tactile means. The power supply shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock. - This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the power supply subassembly shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that receives the power supply subassembly in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating. - This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating. - This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). While exposed to this test condition the power supply shall be subjected to the tests specified in 4.5.2.6 to determine compliance with 3.4.6 as required by 3.3.4.1.

4.5.1.4 Salt spray. - The power supply shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the power supply shall be inspected for any physical damage, then tested in accordance with 4.5.2.1 to 4.5.2.4 inclusive to determine compliance with 3.4 as required by 3.3.6.

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4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the power supply shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While stabilized at the operating temperatures of 3.3.3.1, the power supply shall be tested in accordance with 4.5.2.5 and 4.5.2.6 to determine compliance with the requirements of 3.4.5 and 3.4.6 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the power supply shall be tested in accordance with 4.5.2 to 4.5.2.4 inclusive to determine compliance with 3.4.1 through 3.4.4 as required by 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5 one of the first 5 power supplies produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance.

4.5.2.1 Transistors.

4.5.2.1.1 Direct current gain.- Connect the test circuit to the power supply as shown in Figure I. The input voltage (V_{CE}) and the output current (I_C) shall be as specified in 3.4.1.1 when measured at the pins designated in Figure I. The base current (I_B) when measured at the pins designated in Figure I shall be 27.1 ma minimum and 60 ma maximum to determine compliance with the requirements of 3.4.1.1.

4.5.2.1.2 Dc leakage current.- Connect the test circuit to the Power Supply as shown in Figure 2 to make the following tests.

4.5.2.1.2.1 Collector to base (-20 Vdc).- With switch S1 open and switch S2 closed apply -2.0 ± 0.2 Vdc to Pin 4B of the transistor. The leakage current (I_{CO}) shall be as specified in 3.4.1.2.1 when measured at Pin 1B.

4.5.2.1.2.2 Collector to base (-55.0 Vdc).- With switch (S1) open and switch (S2) closed apply -55.0 Vdc to Pin 4B of the transistor. The leakage current (I_{CO}) shall be as specified in 3.4.1.2.2 when measured at Pin 1B.

4.5.2.1.2.3 Collector to base (-10 Vdc).- With switch S1 closed and switch 52 open apply -10 ± 1.0 Vdc to the transistor. The leakage current I_{EO} shall be as specified in 3.4.1.2.3 when measured at Pin 1B.

4.5.2.1.3 Dc saturation voltage (V_{CE}) sat.).- The test circuit shall be connected to the power supply as shown in Figure 3. With I_C and I_B as specified in 3.4.1.3 the V_{CE} sat. shall not exceed 0.75 Vdc maximum when measured across Pins 4B and 5A.

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4.5.2.2 Diodes.

4.5.2.2.1 Dc forward voltage.- With the forward current (I_F) as specified in 3.4.2.1 the forward voltage drop (V_F) across the diodes shall be measured from the plus terminals to the minus terminals as designated in Table V. The forward voltage drop shall not exceed the maximum requirement as specified in 3.4.2.1

TABLE V

DC FORWARD VOLTAGE TEST

DIODES	TERMINAL NUMBERS CURRENT IN (+)	CURRENT OUT (-)
CR 1	36A, 37A	25A, 26A
CR 5	38A, 39A	25A, 26A
CR 9	40A, 41A	25A, 26A
CR 2	30A, 31A	36A, 37A
CR 6	30A, 31A	38A, 39A
CR 10	30A, 31A	40A, 41A
CR 3	52B, 54B	44B, 45B
CR 7	60B, 61B	44B, 45B
CR 11	63B, 64B	44B, 45B
CR 4	15B, 17B	52B, 54B
CR 8	15B, 17B	60B, 61B
CR 12	15B, 17B	63B, 64B

4.5.2.2.2 Dc leakage current.- With the test conditions as noted in Table VI the leakage current shall be measured through each diode at the pin terminals designated in Table VI. The leakage current shall not exceed the value specified in 3.4.2.2.

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TABLE VI

DC leakage Current Test

TEST CONDITIONS				OPERATING TEMPERATURES		
TERMINAL NUMBERS			VOLTAGE V_R	LOW	NOMINAL	HIGH
DIODES	VOLTAGE (+)	VOLTAGE (-)	(volts dc)	MAX. CURRENT I_B (ua dc)	MAX. CURRENT I_B (ua dc)	MAX. CURRENT I_B (ua dc)
CR 1	25A, 26A	36A, 37A	350 ± 20	0.4	0.4	4.0
CR 5	25A, 26A	38A, 39A	350 ± 20	0.4	0.4	4.0
CR 9	25A, 26A	40A, 41A	350 ± 20	0.4	0.4	4.0
CR 2	36A, 37A	30A, 31A	350 ± 20	0.4	0.4	4.0
CR 6	38A, 39A	30A, 31A	350 ± 20	0.4	0.4	4.0
CR 10	40A, 41A	30A, 31A	350 ± 20	0.4	0.4	4.0
CR 3	44B, 45B	52B, 54B	350 ± 20	0.4	0.4	4.0
CR 7	44B, 45B	60B, 61B	350 ± 20	0.4	0.4	4.0
CR 11	44B, 45B	63B, 64B	350 ± 20	0.4	0.4	4.0
CR 4	52B, 54B	15B, 17B	350 ± 20	0.4	0.4	4.0
CR 8	60B, 61B	15B, 17B	350 ± 20	0.4	0.4	4.0
CR 12	63B, 64B	15B, 17B	350 ± 20	0.4	0.4	4.0

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4.5.2.3 Capacitors.

4.5.2.3.1 DC leakage current.- The Dc leakage current shall be measured after an electrification period sufficient to achieve a stable leakage current value but in no case to exceed 5.1 minutes after application of the dc voltage to the capacitor under test. With the test circuit in Figure 4 connected in succession to the terminals designated in Table VII and with the specified voltage applied, the maximum leakage current shall be as specified in Table I in accordance with 3.4.3.1.

TABLE VII

DC Leakage Current Test

TEST CONDITIONS

CAP.	TERMINAL NUMBERS VOLTAGE (+)	TERMINAL NUMBERS VOLTAGE (-)	VOLTAGE V_C Vdc (volts dc)	VOLTAGE V_C Vac (volts ac rms)
C1	63A	41B	20.0 ± 1.0	0
C2	64A	40B	20.0 ± 1.0	0
C3	65A	39B	60.0 ± 3.0	0
C4	23A	3A	60.0 ± 3.0	0
C5	24A	2A	60.0 ± 3.0	0
C6	29A	1A	60.0 ± 3.0	0
C15	43A	4A	60.0 ± 3.0	0
C7	56B	28B	60.0 ± 3.0	0
C8	61A	51B	60.0 ± 3.0	0
C9	62A	50B	60.0 ± 3.0	0
C14	57B	14B	10.0 ± 0.5	0
C13	58B	13B	10.0 ± 0.5	0

4.5.2.3.2 AC capacitance.- Measure the capacitance at the terminals noted in Table VIII. The capacitance values shall be within the range shown in Table II, column labeled "Nominal Operating Temperatures". Capacitance shall be applied between the terminals as shown in Figure 4 to determine compliance with 3.4.3.2.

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TABLE VIII

TEST CONDITIONS

Capacitor	Terminal Numbers		Voltage V_C		
	Voltage (+)	Voltage (-)	Vdc (volts dc)	Vac. Max. (ac volts rms)	Voltage Frequency (cps)
C1	63A	41B	10.0 ± 0.5	1.02	120 ± 5
C2	64A	40B	10.0 ± 0.5	1.02	120 ± 5
C3	65A	39B	10.0 ± 0.5	1.02	120 ± 5
C4	23A	3A	10.0 ± 0.5	1.02	120 ± 5
C5	24A	2A	10.0 ± 0.5	1.02	120 ± 5
C6	29A	1A	10.0 ± 0.5	1.02	120 ± 5
C7	56B	28B	10.0 ± 0.5	1.02	120 ± 5
C8	61B	51B	10.0 ± 0.5	1.02	120 ± 5
C9	62A	50B	10.0 ± 0.5	1.02	120 ± 5
C10	65B	62B	10.0 ± 0.5	1.02	120 ± 5
C11	60A	59B	10.0 ± 0.5	1.02	120 ± 5
C12	49B	43B	10.0 ± 0.5	1.02	120 ± 5
C13	58B	13B	10.0 ± 0.5	1.02	120 ± 5
C14	57B	14B	10.0 ± 0.5	1.02	120 ± 5
C15	43A	4A	10.0 ± 0.5	1.02	120 ± 5

4.5.2.4 Resistor.- Connect terminals 19B, 20B, 22B, 23B, 24B, 25B and 26B together and connect terminals 29B, 30B, 31B, 32B, 33B, 34B and 35B together. Measure the DC resistance (Item 5) from terminals 19B to 29B and verify that it is 0.1 ohms ± 15% in compliance with 3.4.4. The test current used to measure this resistance shall be no greater than 1.1 amps dc.

4.5.2.5 Low operating temperatures (-40°F).

4.5.2.5.1 Transistor dc current gain.- Repeat the test as specified in 4.5.2.1.1. The base current shall measure a minimum of 27.1 ma and a maximum of 70.5 ma to determine compliance with the requirements of 3.4.5.1.1.

4.5.2.5.2 Diodes.- dc leakage current.- Repeat the test as specified in 4.5.2.2.2. The leakage current shall not exceed the value shown in Table VI under column labeled "Low Operating temperature" in compliance with requirements of 3.4.5.2.1.

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4.5.2.5.3 Capacitors - ac capacitance.- Repeat the test as specified in 4.5.2.3.2. The capacitance values shall be within the range shown in Table II under column labeled "Low Operating Temperatures" to determine compliance with the requirements of 3.4.5.3.1.

4.5.2.6 High operating temperatures (+135°F).

4.5.2.6.1 Transistors - dc leakage current.- Repeat the test specified in 4.5.2.1.2.2. The leakage current shall be as specified in 3.4.6.1.1.

4.5.2.6.2 Diodes - dc leakage current.- Repeat the test as specified in 4.5.2.2.2. The leakage current shall not exceed the value shown in Table VI under column labeled "High Operating Temperature" in compliance with the requirements of 3.4.6.2.1.

4.5.2.6.3 Capacitors - ac capacitance.- Repeat the test as specified in 4.5.2.3.2. The capacitance values shall be within the range shown in Table II under column labeled "High Operating Temperatures" to determine compliance with the requirements of 3.4.6.3.1.

TABLE IX

DC VOLTAGE SUPPLIES

DC SUPPLY NOMINAL VOLTAGE (volts DC)	VOLTAGE TOLERANCE (volts DC)	MAX. RIPPLE (AC VOLTS RMS)	MAXIMUM CURRENT REQUIREMENTS (amps DC)
1.00	± 0.05	0.01	4.3
2.0	± 0.11	0.02	0.01
10.0	± 0.5	0.01	10.5
55.0	± 2.0	0.30	0.01
350	± 20	5.0	0.00001
20.0	± 1.0	0.2	0.00015
60.0	± 3.0	0.6	0.00015

TABLE X

DC CURRENT SUPPLIES

DC SUPPLY NOMINAL CURRENT (ma dc)	CURRENT TOLERANCE (ma dc)	MAX. RIPPLE (AC ma rms)	MAXIMUM VOLTAGE REQUIREMENTS (volts dc)
+100	± 10	± 2	1.2 v

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TABLE XI

AC SIGNAL SUPPLIES

AC SUPPLY MAX. VOLTAGE (ac voltage rms)	FREQUENCY (cps)	MAXIMUM CURRENT REQUIREMENTS (ac ma rms)
1.00 ± 0.02	120 ± 5	750
10.0 ± 0.2	1000 ± 100	15

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.- This assembly consists of two full wave diode rectifier bridges, various capacitors, a power transistor, and a power resistor used in the power supply circuits of the computer. Each rectifier bridge uses six C9144277 type diodes to supply full wave rectified sine wave voltages to the +155V regulator and the -18V PWM regulator. The Power Transistor, 10525749, is used as a shunt regulating element in the -1.5V Regulator. The capacitors are used throughout the power supply circuits as filter capacitors, blocking capacitors, and by-pass capacitors.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable packaging data sheet (see 5.1).
(d) Applicable stock number.
(e) A block of serial numbers to identify each Power Supply Subassembly.

Custodian:

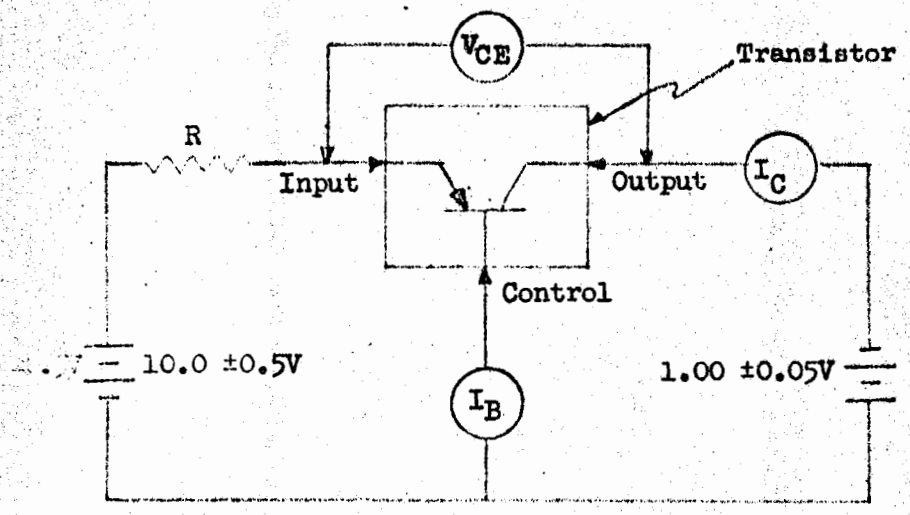
Army - MU

Preparing activity:

Army - MU

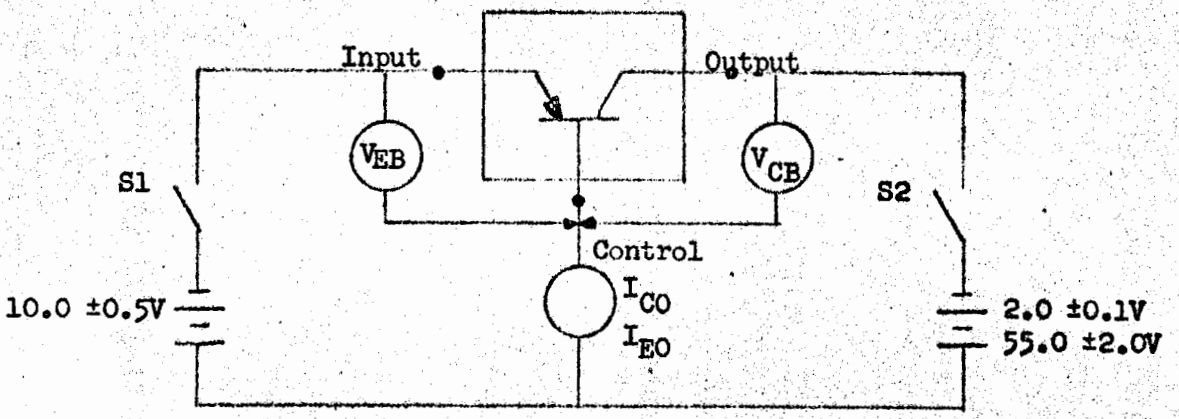
Project No. 1220-A176Y

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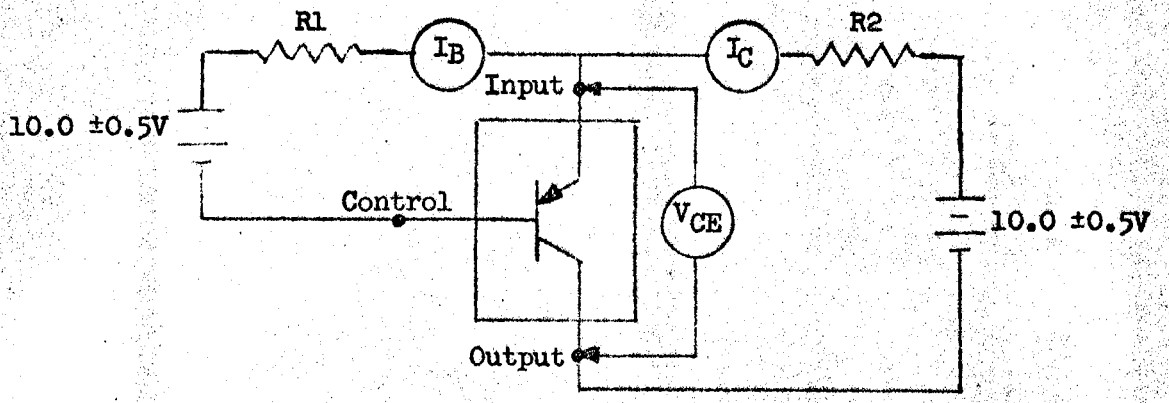
Terminal Numbers		
Input	Control	Output
4B, 5B, 6B 7B, 8B, 10B 11B, 12B	1B, 2B, 3B	5A, 6A, 7A 8A, 10A, 11A 12A, 13A

DC CURRENT GAIN TEST
FIGURE 1



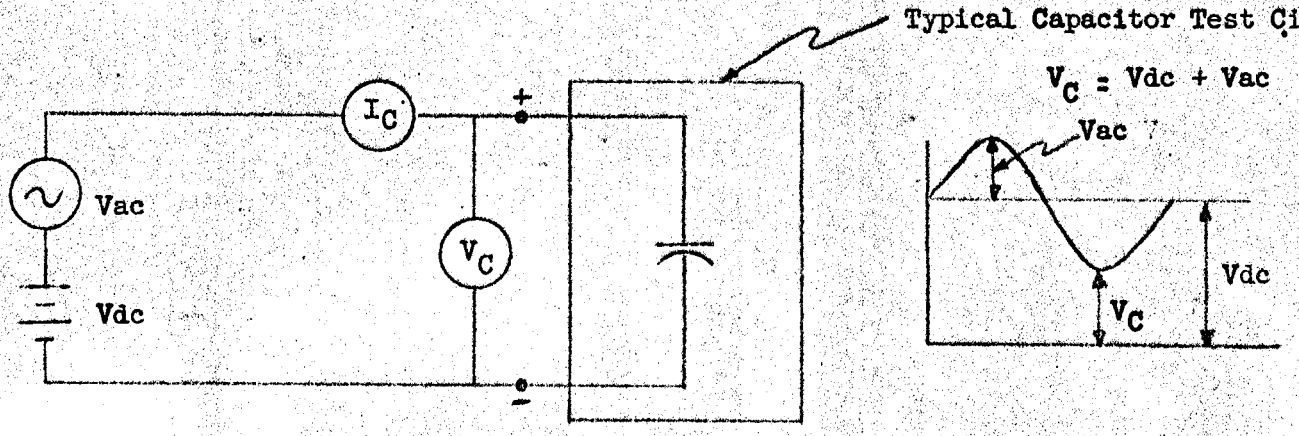
DC LEAKAGE CURRENT TEST
FIGURE 2

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SATURATION VOLTAGE (V_{CEsat}) TEST

FIGURE 3



DC LEAKAGE AND AC CAPACITANCE TEST

FIGURE 4

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